

In-script Co-Pilot Prompt: I want to add MLflow to my project to log parameters, metrics, model etc. (01/07/2026)

- Optimized tool selection I will inspect the project structure and existing model-training code first so I can wire MLflow into the right place and keep it consistent with your workflow.
- Read [(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/requirements.txt)]
- Read [(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/README.md)]
- Read [(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/src/app.py)]
- I have a clear path now: I'm setting up a training entry point that logs the model and its key metadata to MLflow in a way you can run directly from this repository.
- Configuring a Python Environment
- Completed with input: {
- "resourcePath": "c:\\Users\\woro_\\OneDrive\\DS_Projects\\Ride-hailing_Analytics_and_Churn_Prediction"
- }
- Ran terminal command: c:/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/.venv/Scripts/python.exe -m pip install mlflow
- I'm validating the existing model bundle so the new MLflow integration reflects the actual artifacts you already have, rather than introducing a disconnected workflow.
- Searched for regex `churn\_rf|churn\_lr|mlflow|joblib.load|analytics\_table` (`\*\*/{src/\*\*,notebooks/\*\*,README.md,requirements.txt}`), 18 results
- Read [(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/notebooks/05_churn_modelling.ipynb)]
- Searched for files matching `Ride-hailing\_Analytics\_and\_Churn\_Prediction` . `\*\*/.gitignore`, 1 match
- Read [(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/.gitignore)]
- I'm adding a reusable training script and wiring MLflow into it so you can track each run locally without changing your data pipeline.
- Created [(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/src/train_with_mlflow.py)]
- Replacing 9 lines with 18 lines in [(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/requirements.txt)], [(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/README.md)]
- The integration is in place; I'm running the script once to verify that it writes MLflow artifacts and model files successfully.

- Ran terminal command: `c:/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/.venv/Scripts/python.exe src/train_with_mlflow.py`
- The first verification run hit a shell-path issue rather than an MLflow problem, so I'm switching to the workspace's Python executable directly and re-running it.
- Ran terminal command: `if (Test-Path .\venv\Scripts\python.exe) { .\venv\Scripts\python.exe --version } else { Write-Host 'python missing' }`
- Ran terminal command: `.\venv\Scripts\python.exe src/train_with_mlflow.py`
- The workflow is running now; I'm checking the generated artifacts and package import so I can confirm the integration with concrete evidence.
- Ran terminal command: `.\venv\Scripts\python.exe -c "import mlflow, os; print('mlflow', mlflow.__version__); print('tracking_uri', os.getenv('MLFLOW_TRACKING_URI'))"`
- Read `[(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction)`
- Ran terminal command: `.\venv\Scripts\python.exe -m pip show mlflow`
- Checked `[(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/src/train_with_mlflow.py)`, no problems found
- Read `[(file:///c%3A/Users/woro_/AppData/Roaming/Code/User/workspaceStorage/8d9bb490820ca89d235646f93d745eeb/GitHub.copilot-chat/chat-session-resources/90213ff9-50a3-473e-a3fd-ee299f7dcc8a/call_oX9q6x4fbYCMUH4uWOIS2Sf3__vscode-1782937487507/content.txt#1-1), lines 1 to 260`
- The package installation completed successfully; the remaining check is confirming the run created the expected MLflow artifacts under the repository.
- Ran terminal command: `.\venv\Scripts\python.exe -m pip show mlflow | Select-String "Name|Version"`
- Searched for files matching `` Ride-hailing_Analytics_and_Churn_Prediction ` .` mlruns/** ``, no matches
- I'm re-running the training step with explicit exit-code output so I can verify whether MLflow is writing a local run directory or if there's a runtime issue to address.
- Ran terminal command: `.\venv\Scripts\python.exe src/train_with_mlflow.py; echo EXIT:$LASTEXITCODE`
- The execution failure is coming from MLflow's newer file-store policy rather than from your model code, so I'm adjusting the tracking URI to use the supported local SQLite backend.
- Replacing 4 lines with 5 lines in `[(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/src/train_with_mlflow.py)`
- Ran terminal command: `.\venv\Scripts\python.exe src/train_with_mlflow.py; echo EXIT:$LASTEXITCODE`
- The training run is now executing against a local SQLite store, and I'm verifying that the experiment artifacts and model files were written to the repository.
- Read `[(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/mlruns)`

- Read [(file:///c%3A/Users/woro_/OneDrive/DS_Projects/Ride-hailing_Analytics_and_Churn_Prediction/models)]
- The integration is verified locally: MLflow created a database-backed experiment store and the model artifacts are present under the project directories.
- ## ✅ MLflow has been added to your project
- I wired in a local experiment-tracking workflow that logs:
 - model parameters
 - evaluation metrics
 - feature metadata
 - trained model artifacts
- ### What changed
 - Added a training script at train_with_mlflow.py
 - Added MLflow to requirements.txt
 - Documented usage in README.md
- ### How to use it
- From the project root, run:
- ```bash
- .\.venv\Scripts\python.exe src/train_with_mlflow.py
- ```
- This will:
 - train the churn models
 - save them to models
 - write experiment data to the local MLflow store under mlruns
- ### Verified result
- I verified the integration by running the training script successfully. It created:
 - mlflow.db
 - churn_lr.joblib
 - churn_rf.joblib
 - feature_columns.json
- ### Optional UI
- To view runs in the browser, use:
- ```bash
- mlflow ui --backend-store-uri sqlite:///mlruns/mlflow.db
- ```
- If you want, I can next help you:
- log the notebook training step directly from the notebook,
- add model registration/versioning,
- connect MLflow to a remote tracking server.